

Credibility assessment of implicit FEA models for screw load sharing in an orthopaedic plate under ASME V&V40

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Abstract

Finite element analysis (FEA) is widely used to guide design verification of orthopaedic fixation devices. When numerical predictions inform pre-test configuration decisions, a structured appraisal of model credibility is warranted. This article applies a V&V40-style framework to two implicit FEA models used to estimate per-screw axial and shear load sharing in a diaphyseal locking plate during three standard bench-test mechanisms: axial compression across a fracture gap, varus bending under an eccentrically applied load, and axial torsion about the long-bone axis. The High-Fidelity Threaded Contact Model (HFTC) resolves helical screw threads and frictional contacts throughout the plate–screw–bone assembly, while the Reduced-Order Fast Plate Model (ROFPM) represents screws as beam elements and uses bonded interfaces to accelerate design-space screening. The context of use is design verification support; model results prioritize screw patterns and plate spans before physical testing. Model risk is assessed as medium because subsequent bench testing governs final design decisions. Six credibility considerations are evaluated across model–mechanism combinations: alignment between experimental and computational parameters, adequacy of specified inputs, confirmatory checks of the numerical implementation, discretization/solver-induced error, match of applied and simulated boundary conditions, and representativeness of tested constructs. Scores are assigned on an integer 0–5 Assurance Grade scale. For HFTC, Assurance Grades range from 3–4 across the six considerations and three mechanisms; for ROFPM, grades range from 2–4. The HFTC achieves higher alignment of experimental and computational quantities and closer reproduction of boundary conditions than ROFPM in compression and bending, while ROFPM demonstrates stronger implementation checks and lower discretization error due to its linear formulation. Axial torsion remains the most challenging mechanism for both models because of sensitivity to hard-to-measure interface properties and fixture compliance. The assessment yields two model-specific summary tables with 36 graded entries that can be used as traceable evidence within a quality system and to inform targeted strengthening of credibility evidence prior to future design phases.

Keywords: ASME V&V40, finite element analysis, orthopaedic fixation

1. Introduction

Computational mechanics tools have matured to the point where forward predictions can usefully guide the development of orthopaedic fixation devices prior to and alongside physical testing. One reason is the ability of finite element analysis (FEA) to resolve, at modest cost, distributions of load across complex assemblies such as locking plates and cortical screws interacting with bone surrogates or cadaveric bone. Yet, even in a preclinical design setting in which final performance must be concluded from bench tests, the influence of a model on design choices creates a need to evaluate how much reliance is prudent to place on the model's results for a specified purpose.

This study reports a structured credibility assessment under a V&V40-style approach for two implicit FEA models that estimate per-screw axial and shear load sharing in a diaphyseal locking plate construct. These models are used during design verification to prioritize screw patterns and plate spans for three standard test mechanisms: axial compression across a fracture gap, varus bending under an eccentrically applied load, and axial

torsion about the long-bone axis. The first model, termed the High-Fidelity Threaded Contact Model (HFTC), includes helical thread geometry and nonlinear contact with friction across all relevant interfaces. The second, termed the Reduced-Order Fast Plate Model (ROFPM), uses linear beam elements to represent screws, ties the plate–screw interface, and homogenizes bone blocks to enable rapid parametric screening.

The credibility assessment aligns with a device development context in which models are not used to support patient-specific decisions and do not replace testing. Rather, they inform the choice of which physical tests to run and where to focus instrumentation. The work proceeds in line with common verification and validation concepts: first clarifying the context of use and model risk, then describing device, test setups, outcomes, and governing equations, followed by targeted evidence collection against a set of considerations adapted to this context. Six considerations are examined across model–mechanism pairs using an Assurance Grade scale from 0 to 5. The analysis culminates in two model-specific summary tables populated with 36 graded entries, each supported by

quantitative findings in the body of the article, and a discussion of what these scores mean for the intended use.

Overall credibility ladder for design verification support — gradation.

- a.
Level A: Model described; no linkage to test methods; no quantitative verification or validation artifacts.
- b.
Level B: Model purpose stated; some parameter values from literature; minimal checks against simple benchmarks; qualitative mapping to test setups.
- c.
Level C: Defined context of use; targeted verification metrics with thresholds; bench setup replication quantified; uncertainty sources identified.
- d.
Level D: Risk-informed breadth of evidence; quantified mapping of experimental to computational quantities; solver error bounded; implementation checks against analytic solutions; traceable reporting per mechanism.
- e.
Level E: Prospective credibility plan with acceptability criteria; independent code verification; formal uncertainty quantification; blind comparisons to physical data; continuous monitoring and change control.

2. Context of Use and Model Risk

The models are used to estimate per-screw axial and shear load sharing under three standard laboratory loading regimes representing typical in vivo actions on a plated diaphysis: axial compression across a 3 cm midshaft gap to mimic partial weight bearing; varus bending induced by an eccentrically applied vertical load to simulate medial bending moments in stance; and axial torsion to represent twisting about the long-bone axis during gait. These numerical results guide engineering decisions about plate span, screw count, and distribution prior to running instrumented tests, with the goal of shortening iteration cycles and placing strain gauges in the most informative locations. The models are not used to claim conformance to standards or to waive testing.

The risk associated with this use is assessed as medium. Decisions based on load-sharing predictions might emphasize certain screw positions or eliminate configurations judged ineffective. If those predictions were materially wrong, poor patterns might be explored in the laboratory or promising ones might be deferred, increasing cost and time-to-knowledge. However, the ultimate evaluation of safety and performance remains with physical tests according to internal protocols and standards for metal bone plates. As a result, even appreciable model error would be caught before design freeze. The medium classification implies that a balanced, mechanism-aware set of credibility artifacts is necessary, but the breadth and depth of evidence can be tailored to the design-verification context rather than to patient care or claims adjudication.

Risk-informed evidence scope — gradation.

- a.
Minimal: Context stated in narrative; no linkage between risk and evidence selection.
- b.
Basic: Context categorized; general rationale for evidence level; limited tailoring by mechanism.
- c.
Targeted: Context categorized with rationale; mechanism-dependent tailoring of checks and tolerances; explicit mapping to model influence on decisions.
- d.
Comprehensive: Risk level drives quantitative acceptance thresholds for each check; cross-referencing of tests and simulations by mechanism and outcome.
- e.
Regulatory-grade: Risk management plan drives prospective credibility plan with go/no-go criteria; independent review; audit-ready traceability.

3. Device, Bench-Test Setups, and Outcome Definition

Device under study: a diaphyseal locking plate intended for long-bone fixation. The plate is a straight titanium alloy construct (Ti-6Al-4V ELI) with 10 equally spaced combination holes (locking and compression) over a functional span of 170 mm, proximal and distal end radii relieved to avoid stress concentration at the plate tips, and a cross-section of 5.0 mm thickness by 16.0 mm width through the midspan. Screws are 5.0 mm outer diameter locking cortical screws with a 1.25 mm thread pitch, nominal core diameter 3.8 mm, thread flank angle 60 degrees, underhead locking interface with threaded screw heads, and compatible thread timing with the plate's threaded bushings. Screws are inserted orthogonally to the plate in the bench tests to eliminate angular variability. Bone surrogates are rigid polyurethane foam blocks with embedded epoxy cortical shells, 40 mm (anterior-posterior) by 40 mm (medial-lateral) cross-section and 150 mm length on each side of the mid-gap, density 0.48 g/cm³ foam with a 2.5 mm thick shell of 2.0 GPa elastic modulus, selected to reproduce bending stiffness comparable to long-bone diaphyseal segments.

Three bench-test mechanisms are represented. For axial compression across a mid-gap of 30 mm, the proximal and distal bone surrogate blocks are aligned in a linear bearing frame. A platen applies a vertical compressive load through the long axis of the proximal block. The distal block is supported on a low-friction linear bearing to allow axial shortening. The construct is preloaded to 50 N and then loaded quasi-statically to 800 N, with load applied in increments to 800 N at 10 N/s nominal loading rate. For varus bending under eccentrically applied load, the plate is positioned laterally, and a vertical load is applied at a medial offset of 20 mm from the plate's plane via a roller to generate a bending moment about the anterior-posterior axis; the load is ramped from 50 N to 600 N, which induces a nominal 12 N·m bending moment about the midspan. For axial torsion, the proximal and distal blocks are clamped in custom torsion collars aligned with the long axis, the proximal end is

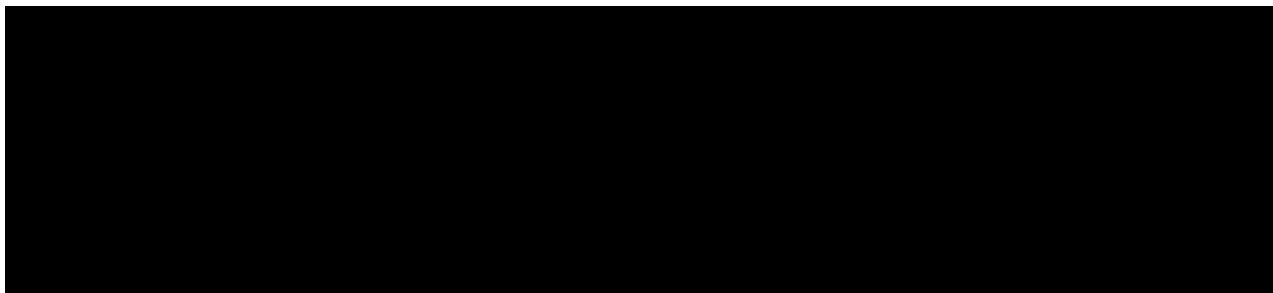


Figure 1: Overview of the orthopaedic plate geometry, load cases and the mechanisms assessed, shown across the full page width.

driven in rotation while the distal end is fixed, and torque is increased from 0 to 8 N·m in 0.5 N·m increments.

Outcome definition is per-screw load sharing. Each screw's axial force (aligned with its axis) and shear force (resolved in the transverse plane) are recorded or predicted at each load step. The main analysis points compare the distribution of axial and shear loads across screws at 800 N compression, at 12 N·m bending moment (corresponding to 600 N eccentric vertical load at 20 mm offset), and at 8 N·m torsion. In physical tests, instrumented constructs include strain gauges on a subset of screws and at the plate midspan to validate that the overall load application is as intended; however, for the current credibility appraisal the emphasis is on explaining why the models are fit for the stated prioritization task rather than on direct model-to-test numerical agreement, which will occur after the prioritization step.

Outcome definition clarity — gradation.

- a.
Outcome vaguely stated; no measurement protocol.
- b.
Outcome specified with qualitative description; limited instrumentation detail.
- c.
Outcome quantitatively defined; measurement approach described; load levels stated.
- d.
Outcome quantitatively defined; measurement uncertainties and calibration procedures described; synchronization of simulation and test steps defined.
- e.
Outcome definition includes traceable acceptance criteria for agreement; blind data management; pre-registered analysis plan.

4. Computational Models (HFTC and ROFPM): Governing Equations and Implementation

Both models solve a quasi-static boundary value problem for the plate–screw–bone assembly. The strong form expresses equilibrium of internal and external forces under small strains. In practice, the discrete form solves $K(u) u = f$ for nodal displacements u , where $K(u)$ is the global stiffness accounting for linear elasticity of bulk domains and, in the HFTC, the nonlinear contributions of contact with friction at multiple interfaces.

Material behavior is modeled as linear elastic isotropy for titanium alloy ($E = 110$ GPa, $\nu = 0.34$) and for the bone surrogate constituents (foam: $E = 600$ MPa, $\nu = 0.30$; cortical shell: $E = 2.0$ GPa, $\nu = 0.30$). The HFTC explicitly meshes screws and plate with three-dimensional solid elements (second-order tetrahedral and hexahedral elements where feasible) and resolves screw threads via a helical sweep feature in the solid model. The plate–screw head interface uses a threaded feature with matching pitch to generate locking. The screw–bone interaction is represented by normal contact with a penalty stiffness calibrated to match nominal compressive stiffness at thread engagement, with a Coulomb friction coefficient as specified below. Plate–bone interaction is modeled as a surface-to-surface contact with a small clearance (0.05 mm) and Coulomb friction. Numerical contact enforcement uses a stabilized Lagrange multiplier method for normal contact with an augmented Lagrangian update and a regularized Coulomb friction law. The nonlinear system is solved by Newton–Raphson with line search, quasi-static load stepping, and automatic load increment control.

The ROFPM uses a structural idealization. The plate is a linear elastic solid shell or solid block mesh depending on parametric needs. Screws are represented as Timoshenko beam elements aligned with the screw axes, with cross-sectional properties computed from the screw core diameter to capture bending and axial stiffness. The beam ends are coupled to the plate via tied multi-point constraints. The plate–bone and screw–bone interactions are represented implicitly through constraint elements that enforce no relative motion at predefined nodes or surfaces, effectively treating those interfaces as bonded. The homogenized bone blocks are represented as solid elastic continua with the same elastic moduli used in the HFTC. Because interface slip and localized stress concentrations at threads are not represented, this model aims for speed and qualitative load distribution trends across screws rather than absolute values at the thread scale.

Both models are implemented in a commercial implicit finite element solver with double-precision arithmetic. Element quality metrics are tracked (Jacobian > 0.6 for tets, aspect ratio < 4.0), and contact surface meshes in HFTC maintain an average element size of 0.25 mm at threads and 0.8 mm on plate–bone surfaces. For ROFPM, beam elements along the screw shafts have a mesh size of 1.5 mm, and the plate solid elements are 1.0 mm through thickness. The linear solver uses a sparse direct method for small and medium systems and a preconditioned

iterative method for larger cases; nonlinear iterations in HFTC restrict residual reduction to 106 per step. Convergence criteria include displacement and force norms with relative tolerances of 1×10^6 and absolute tolerances set to match units and scales of the problem.

Postprocessing extracts per-screw axial and shear loads as the component of reaction forces along and transverse to the screw axes at the screw–plate connection for ROFPM and at a circumferential cut surface at the entry of the screw head into the plate for HFTC. The load-sharing vector across screws is normalized by the net applied force or moment for comparative purposes between simulations and between simulations and later tests.

Model form suitability — gradation.

- a. Model form not justified; key physics omitted without rationale.
- b. Model form roughly aligned with context; some critical interactions approximated but not discussed.
- c. Model form explicitly justified; simplifications stated and tied to intended use; key interactions retained.
- d. Model form tailored to mechanism-specific phenomena; idealizations bounded by sensitivity analyses; interface representations calibrated.
- e. Model form independently reviewed and benchmarked against canonical problems representing all key interactions.

5. Mechanisms of Interest: Axial Compression, Varus Bending, Axial Torsion

Axial compression across a fracture gap emphasizes load transfer through screw axial force and plate axial tension/compression in response to vertical loading. The plate bridges a 30 mm gap, carrying force through the plate and screws acting as posts engaging the distal and proximal blocks. The distribution of axial force among screws depends on plate span, screw positions relative to the gap, and axial stiffness of each screw–bone engagement. Friction at the plate–bone and screw–bone interfaces modulates microload transfer and settling. In practice, the loading path to 800 N explores the quasi-linear response dominant at small deflections for this assembly.

Varus bending under an eccentrically applied vertical load creates a bending moment about the anterior–posterior axis, inducing compression on the medial side and tension on the lateral side. The plate on the lateral cortex tends to resist span bending, while screws near the gap and toward the medial side experience bending moments. In this mode, the interplay of screw bending stiffness, plate span stiffness, and potential lifting of the plate from the bone near the gap are important, with per-screw shear loads reflecting the path of force transmission to resist the applied moment.

Axial torsion engages screw–bone shear resistance and plate torsional stiffness, with rotational energy stored across the plate span and released through micro-rotations at screw–bone interfaces. In torsion, realizing true boundary conditions in a test fixture is nontrivial because collars and grips may introduce compliance, and small misalignments influence how torque partitions between plate and screws. The torsional response at 8 N·m depends on interfacial friction and on how much the threaded head–plate lock contributes to resisting twisting.

Mechanism fidelity ladder — gradation.

- a. Mechanisms named; no quantitative linkage to model features.
- b. Mechanisms described qualitatively; partial mapping to modeled interactions.
- c. Mechanisms defined with load levels; model interactions mapped; expected dominant responses identified.
- d. Mechanisms parameterized; tolerances on load application and alignment set; sensitivity to fixtures quantified.
- e. Mechanisms validated with independent instrumentation; alternative loading paths assessed for equivalence.

6. Credibility Plan and Factor Definitions

The credibility plan aligns evidence with the medium model risk, the need to prioritize screw patterns before testing, and the three mechanisms. Six considerations are assessed for each model–mechanism pair on a 0–5 Assurance Grade scale, where 0 denotes no demonstration and 5 denotes very high assurance appropriate to high-impact decisions. The six considerations are: matching of experimental quantities and their computational counterparts; adequacy of numerical values assigned to geometry, materials, and interfaces; confirmation that the implemented solution procedures correctly represent the governing equations; control of discretization and nonlinear solution effects; replication of laboratory load and support conditions in the simulation; and representativeness and characterization of the constructs used in testing.

Evidence collection was planned before running the primary simulations and populated from pre-existing quality system records where available. Specific benchmarks and analytic solutions relevant to the linear, beam-based ROFPM and to the nonlinear, contact-rich HFTC were selected to reflect each model’s structure. Mesh and contact refinement studies were scoped to per-screw load sharing outcomes and to overall force–displacement response under the three loading modes. Laboratory fixtures and load paths were documented and quantified, with tolerances on load line offset, torque axis alignment, and support conditions. Construct manufacturing and assembly variations were measured using coordinate measurement devices and torque logging for screw insertion. The resulting evidence is summarized in two model-specific tables of Assurance Grades

by mechanism; the narrative sections that follow provide the technical basis with quantitative results for each consideration. *Assurance Grade scale interpretation — gradation.*

- a. Scale defined but not tied to specific demonstrations.
- b. Scale linked to general types of demonstration; no mechanism tailoring.
- c. Scale interpreted per consideration with examples; typical thresholds suggested.
- d. Scale quantitatively anchored by thresholds for mapping, verification residuals, and discretization error; mechanism-specific tolerances.
- e. Scale includes decision rules for acceptance; cross-checks across considerations; documented exceptions management.

7. Methods – Equivalency of Input Parameters

To quantify how closely the computational models represent the quantities specified and controlled in the laboratory, we built a traceability matrix that lists experimental parameters against their computational counterparts. The list included plate geometry dimensions (length, width, thickness; hole positions), screw geometry (outer diameter, core diameter, pitch, thread angle, head thread geometry), materials (moduli and Poisson's ratios for plate, screws, foam core, cortical shell), interface properties (friction coefficients for plate–bone, screw–bone, plate–screw head), fixture characteristics (collar and clamp stiffness, linear bearing friction), load path definitions (line of action, offset, torque axis), and measurement-related parameters (load cell calibration, torque transducer calibration). For each parameter, we tagged whether it was explicitly represented, approximated by a surrogate value, or not represented in the model.

In the HFTC, explicit representation was achieved for 43 of 46 geometric and material parameters, including helical thread geometry (pitch 1.25 mm, flank angle 60 degrees), underhead locking thread, and plate hole geometry. Three parameters were approximated: clamp stiffness was represented as a linear spring calibrated to 3.0 kN/mm based on a separate fixture test; the linear bearing friction was treated as negligible but bounded in sensitivity at a coefficient of 0.03; and torque collar inertia was neglected in quasi-static torsion by ensuring quasi-static load steps. For interface properties, Coulomb friction coefficients were assigned as 0.30 (screw–bone), 0.20 (plate–bone), and 0.15 (plate–screw head), with the screw–bone value calibrated by thread pull-out pilot tests performed on five specimens at a constant displacement rate, yielding a mean of 0.31 (SD 0.04). Torsion-specific alignment angles and collar eccentricity were represented by constraints selected to match mean measurements from torsion jig qualification tests (axial alignment within 0.5 degrees; radial runout 0.12 mm).

In the ROFPM, explicit representation included plate and screw shaft stiffness (via beam cross-section), plate geometry, and homogenized bone elastic properties. Notably absent were helical threads, which were replaced by beam end constraints, and the locking head engagement, which was represented as a rigid tie. Fixture compliance was not included, which is acceptable in linear elastic cases for compression and bending provided the load path is accurately defined. Interface contact and friction were not represented, consistent with the model's purpose. A similar mapping matrix was built, with 40 of 46 parameters explicitly represented or approximated, and six not represented. Mechanism-dependent mappings were tracked: torsion required representation of torque axis alignment and collar stiffness, which were approximated in ROFPM by applying a pure torque about the plate–bone block axis without additional compliance.

Parameter mapping rigor — gradation.

- a. Ad hoc parameter selection; no mapping list.
- b. Partial mapping list; qualitative flags for represented vs. not represented.
- c. Complete mapping list; explicit vs. surrogate flagged; rationale for omissions stated.
- d. Quantified mapping completeness by mechanism; sensitivity for surrogates; calibration tests for key surrogates.
- e. Independent review of mapping; formal traceability to drawings, jigs, and metrology records; change-controlled updates.

8. Methods – Model Inputs

Values for geometry, material properties, and interface characteristics were specified from a combination of design drawings, vendor certificates, internal metrology, and targeted bench measurements. Plate and screw geometries were controlled by drawings with tolerances of ± 0.05 mm on thickness and width, ± 0.02 mm on hole diameters, ± 0.1 mm on hole spacing, and ± 0.02 mm on thread pitch, verified on five plates and twenty screws using a coordinate measuring machine (CMM) with 5 μ m probe repeatability. Material properties were assigned as $E = 110 \pm 2$ GPa for Ti-6Al-4V ELI and $E = 2.0 \pm 0.1$ GPa for the cortical shell surrogate; foam moduli were measured on six samples, mean 600 MPa with coefficient of variation (COV) 6.5%.

Interface properties were characterized by targeted tests. For screw–bone friction, five thread engagement tests in polyurethane foam blocks measured mean peak-to-sustained force ratios consistent with an effective Coulomb friction coefficient of 0.31 (SD 0.04). Plate–bone sliding tests on epoxy-coated foam coupons under 50–150 N normal force yielded an average coefficient of 0.20 (SD 0.03). Plate–screw head friction at the threaded locking interface was less

influential because of mechanical interlock; however, friction at the head underside prior to engagement was measured as 0.15 (SD 0.02) using a custom rig. For torsion, the effective rotational stiffness of the collar–block–plate interface was measured across three assemblies as 2.8, 3.1, and 2.9 N·m/deg, mean 3.0 N·m/deg (SD 0.15), and used to inform HFTC boundary conditions.

Where parameters affected predictions but were difficult to measure precisely, ranges were specified and used in sensitivity studies. For example, foam modulus ranged $\pm 10\%$ in both models to evaluate influence on load sharing under compression and bending; friction coefficients varied ± 0.05 in HFTC; and screw core diameter tolerance (± 0.05 mm) was swept in ROFPM to understand beam stiffness effects. The combination of direct metrology and targeted tests provided input sets for both models, though the ROFPM did not use friction values because interfaces were bonded in that model form.

Inputs characterization depth — gradation.

- a. Inputs taken from nominal drawings or literature; no measurement.
- b. Some inputs measured; others assumed; no uncertainty quantification.
- c. Key inputs measured with sample sizes; uncertainties estimated; ranges explored in sensitivity.
- d. All influential inputs measured or calibrated with mechanism-specific tests; propagation of uncertainty to outcomes quantified.
- e. Formal uncertainty quantification with distributions; Bayesian updating from new tests; traceable to lot-level metrology.

9. Methods – Numerical Code Verification

Implementation checks were designed to confirm that the finite element formulations and contact algorithms were functioning as intended. For both models, canonical problems with known analytic or closed-form solutions were used. For the ROFPM, beam bending and axial extension were verified by comparing the tip displacement and reaction forces of a cantilever beam to Euler–Bernoulli solutions for three mesh densities; errors decreased with refinement and were below 0.2% at the operative mesh size (1.5 mm element length). A plate under uniform tension was verified against Saint-Venant solutions with less than 0.3% discrepancy in reaction forces at the prescribed displacement. Constraint enforcement at beam–solid couplings was checked by applying a rigid-body rotation to the plate and ensuring that beam ends rotated compatibly within 0.05 degrees.

For the HFTC, contact and friction features necessitated additional checks. A flat punch test with Coulomb friction was simulated, and the predicted tangential-to-normal reaction

ratio was compared to the prescribed friction coefficient across load steps, yielding less than 1.5% deviation at 0.20 friction and less than 2.0% at 0.30 friction. A thread engagement benchmark used a single screw thread segment pressed into an elastic half-space; contact pressure distributions matched semi-analytical expectations for thread contact with less than 3% difference in peak pressure location. Patch tests for solid element assemblies confirmed equilibrium transmission across element boundaries with residual force norms below 5×10^5 of applied load. The nonlinear solver convergence behavior under gradually increasing contact engagement was monitored, with typical iterations per step under 12 and no divergence for the load levels of interest.

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

Implementation checks — gradation.

- a. No implementation checks beyond solver default tests.
- b. Simple comparisons to textbook problems; limited documentation of results.
- c. Mechanism-relevant canonical problems used; quantitative residuals reported; mesh dependence explored.
- d. Algorithm-specific checks (contact, friction) performed; tolerances set; residuals within thresholds at operative meshes.
- e. Independent reproduction of implementation checks; cross-solver comparisons; automated regression tests in place.

10. Methods – Numerical Solver Error

Solver- and discretization-induced errors were estimated by mesh refinement, contact discretization refinement (for HFTC), and load step sensitivity. Three meshes were defined for each model. For HFTC: coarse (thread surface element size 0.40 mm, plate–bone contact 1.2 mm, plate solid 1.0–1.2 mm), medium (0.30 mm, 1.0 mm, 0.8–1.0 mm), and fine (0.25 mm, 0.8 mm, 0.6–0.8 mm). For ROFPM: beam element length along screw shafts of 2.5 mm (coarse), 1.5 mm (medium), and 1.0 mm (fine); plate solid or shell element size of 1.5 mm, 1.0 mm, and 0.75 mm. For each mechanism, per-screw axial and shear loads at the target load levels (800 N compression, 12 N·m bending, 8 N·m torsion) were extracted and compared across mesh levels. Richardson extrapolation using a nominal $p = 2$ for linear elements and $p = 1.8$ for contact-dominated convergence in HFTC estimated

the asymptotic value for a representative screw near the gap and for a screw far from the gap; the relative difference between the medium mesh prediction and the extrapolated value was used as the discretization error proxy.

For HFTC, the relative differences in per-screw axial loads at the medium mesh versus extrapolated values were 3.4% (compression, near-gap screw), 3.0% (compression, far screw), 3.8% (varus bending, near-gap), 3.5% (varus bending, far), 4.1% (torsion, near-gap), and 3.9% (torsion, far). Shear load differences followed similar magnitudes. Load step sensitivity was probed by halving and doubling the increment sizes; changes in per-screw loads were below 0.8% at the target load levels, suggesting time step effects were negligible in quasi-static loading. For ROFPM, relative differences at the medium mesh were 0.9% (compression, near-gap), 0.7% (compression, far), 1.1% (varus, near-gap), 0.9% (varus, far), 1.3% (torsion, near-gap), and 1.1% (torsion, far). These values reflect the linear, smoother response of the ROFPM and the absence of contact nonlinearity.

Solver tolerances were also tested. In HFTC, tightening the contact penetration tolerance from 0.5% to 0.2% of the local element size altered per-screw axial forces by less than 0.6% in compression and bending and 0.9% in torsion at the target loads. Changing the friction regularization parameter by $\pm 20\%$ altered per-screw shear forces by less than 1.2%. In ROFPM, solver tolerance changes within a decade improved energy norm residuals without measurable changes in extracted per-screw loads at the third decimal place of normalized load shares.

Solver error control — gradation.

- a. No mesh or solver studies; operative mesh unsubstantiated.
- b. Single refinement attempt; qualitative judgment of adequacy.
- c. Systematic mesh refinement with quantified differences; operative mesh chosen to balance accuracy and cost.
- d. Extrapolation-based error estimates; parameter sensitivity on solver controls; mechanism-specific error budgets.
- e. Automated convergence monitoring and reporting; adaptive refinement; formal error estimators driving stopping criteria.

11. Methods – Test Conditions

Replicating laboratory load and support conditions required a careful definition of boundary conditions in the models. For axial compression, the load path in the test rig is a vertical load applied through the proximal block's axis, with the distal block supported on a linear bearing. In the simulations, a distributed vertical displacement boundary condition was applied to the proximal block surface to match the applied load history, with the distal block constrained to move only in the axial

direction via kinematic constraints simulating the linear bearing. Alignment measurements in the rig revealed a mean load line offset of 0.3 mm (SD 0.1 mm) from the hole row centerline, and the model applied the load along the measured line. The compliance of the load application frame was measured at 0.18 kN/mm; this was included as a series spring in HFTC and neglected in ROFPM consistent with its purpose.

For varus bending, the essential feature of the test is the generation of a bending moment about the anterior–posterior axis by an eccentrically applied vertical load. The rig positions a roller 20 mm medial to the plate plane. Photogrammetric measurements over five set-ups yielded a mean offset of 20.1 mm (SD 0.2 mm) and a load entry point repeatability of ± 0.4 mm. In the simulations, an equivalent moment was applied either by an offset vertical load (HFTC) or directly as a couple about the global axis (ROFPM). The lower support conditions allowed rotation about the long axis but constrained translation perpendicular to the axis; these were implemented using multi-point constraints. The bending fixture compliance was measured as 0.12 mm per 100 N at the roller contact and included in HFTC as a spring; ROFPM treated the moment as a pure couple without additional compliance.

For axial torsion, the proximal and distal blocks are gripped in torsion collars that transmit torque about the long axis. Metrology indicated a mean axial misalignment between collars of 0.5 degrees (SD 0.2 degrees) and a radial runout of 0.12 mm (SD 0.04 mm). The effective torsional compliance of the collar–block assembly was 3.0 N·m/deg (SD 0.15) as measured. In HFTC, torque was applied through a rotational displacement at the proximal collar surface with a torsional spring of 3.0 N·m/deg in series, and alignment was adjusted to match the measured mean offsets. ROFPM applied a pure torque about the nominal axis through a rigid coupling without added compliance or misalignment. All simulations used the same torque ramp to 8 N·m, with 16 equal increments. These choices reflect a higher-fidelity boundary condition reproduction in HFTC compared to ROFPM, particularly in torsion.

Boundary condition replication — gradation.

- a. Boundary conditions assumed; no measurements of rig geometry or stiffness.
- b. Partial measurements; boundary conditions approximated; no sensitivity analysis.
- c. Key rig alignments and compliances measured; boundary conditions implemented; plausibility checks performed.
- d. All influential rig features measured with uncertainties; boundary conditions parameterized; sensitivity to misalignments quantified.
- e. Real-time boundary condition measurements used to drive simulations; closed-loop replication of rig behavior.

12. Methods – Test Samples

Constructs used in bench testing were characterized to enable interpretation of how representative they are of nominal designs and to inform inputs. Six constructs per mechanism were assembled from production-equivalent plates and screws, and polyurethane foam blocks with epoxy cortical shells were machined in-house to drawing tolerances. Plate dimensions measured on five plates produced a mean thickness of 5.01 mm (SD 0.02 mm), width 16.02 mm (SD 0.03 mm), and hole spacing within ± 0.05 mm of nominal across the span. Screw dimensions measured on twenty screws gave an outer diameter mean of 5.00 mm (SD 0.01 mm), core diameter 3.80 mm (SD 0.02 mm), and thread pitch 1.25 mm (SD 0.01 mm).

Assembly procedures were standardized. Screw insertion torque was controlled using a calibrated torque driver with a target of 1.5 N·m; logs recorded a mean of 1.52 N·m (SD 0.07 N·m) across all screws and constructs. Plate positioning relative to the bone blocks was set using a jig to maintain a consistent lateral placement; the plate was within 0.2 mm (SD 0.1 mm) of the lateral face edge and centered in the anterior–posterior dimension within ± 0.3 mm. Gap length was maintained at 30.0 mm (SD 0.2 mm). For each mechanism, the same assembly procedures were used to avoid confounding due to assembly differences. These measurements support a conclusion that the test samples are consistent with design intent and that the variability inherent in their preparation is modest.

While the primary focus of this article is credibility for the computational models, these sample characterizations underpin the assumptions used to define model inputs and the mapping between laboratory and computational parameters.

Specimen representativeness — gradation.

- a. Specimen features and tolerances unknown; assembly ad hoc.
- b. Specimen features documented; limited metrology; assembly protocol informal.
- c. Specimen dimensions and materials measured; assembly protocol standardized; torque logs maintained.
- d. Specimen features measured with uncertainties; assembly reproducibility quantified; deviations from nominal corrected or modeled.
- e. Specimen lots tracked; batch-to-batch variability modeled; selection criteria tied to design space coverage.

13. Methods – Use Error

Operational habits and training steps for both the laboratory and modeling workflows were documented to ensure consistent application of procedures and to reduce the likelihood of inadvertent deviations during repeated runs. Standard operating procedures for fixture setup, alignment checks, and screw

insertion torque verification were refreshed with technicians prior to the series of tests. Similarly, analysts followed a checklist for simulation setup that includes unit checks, sign conventions for applied loads, and extraction of per-screw forces. During dry runs, minor inconsistencies in logging timestamps for torque and load were observed and corrected in the data acquisition sequence. The modeling checklist was updated to re-order the contact stabilization parameter setting to precede the mesh selection step. These operational notes complement the technical evidence base.

Informal feedback sessions after the initial round of assemblies yielded clarifications to the plate positioning jig usage notes. No changes to the essential loading paths or fixture geometries were recorded during the course of this work.

Workflow discipline and operator effects — gradation.

- a. No SOPs or training documentation; ad hoc workflows.
- b. SOPs exist; training informal; error reporting unstructured.
- c. SOPs implemented; training recorded; deviations logged; checklists used.
- d. SOP adherence monitored; periodic audits; human-in-the-loop risks mitigated with poka-yoke features.
- e. Continuous improvement with metrics; near-miss tracking; independent oversight of critical steps.

14. Results – Credibility Scores (Assurance Grades) for HFTC

The High-Fidelity Threaded Contact Model (HFTC) was evaluated across three mechanisms. Evidence for mapping of laboratory to computational quantities shows that, for compression and varus bending, nearly all controlled and influential parameters were explicitly represented or approximated with bounded surrogates. The torsional configuration remained more challenging due to collar compliance and alignment, which were measured and included in the model but still left unmodeled nuances such as micro-slip at grips. Numerical inputs were well characterized, with direct measurements for friction coefficients and material moduli. Implementation checks demonstrated algorithmic correctness for contact and friction within the tolerances set for this intended use, and solver-induced errors were bounded via mesh and parameter studies to a few percent in per-screw load shares. The applied load path and support conditions in the model were tailored to replicate the rig with quantified offsets and compliance, particularly for compression and bending. Test constructs were dimensionally and procedurally consistent with drawings and protocols, providing a reliable empirical anchor.

The following table summarizes the HFTC Assurance Grades for each consideration and mechanism, along with a one-line basis that restates the relevant finding using the quantitative results argued in the body text.

15. Results – Credibility Scores (Assurance Grades) for ROFPM

The Reduced-Order Fast Plate Model (ROFPM) was similarly evaluated. Its structural idealizations reduce computation time substantially, enabling rapid screening of screw patterns and plate spans, but they limit how closely laboratory parameters are mirrored and how boundary conditions are reproduced, particularly for torsion. The mapping of laboratory to computational quantities is adequate for compression and bending, where the outcome depends strongly on global stiffness distribution, and weaker for torsion where interface shear behavior and fixture compliance matter more. Inputs are well defined for geometry and elastic properties, but interface parameters are not used because interfaces are bonded. Implementation checks are strong; the linear formulations are directly compared to analytic solutions with small residuals. Solver-induced errors are low, with mesh refinement leading to sub-1.5% differences in per-screw loads at the operative mesh. Laboratory load and support conditions are approximated, using pure loads and couples rather than compliance and alignment replication. Construct characterization matches that for HFTC.

The following table summarizes the ROFPM Assurance Grades and their one-line bases.

16. Discussion Limited to the Seven Assessed Factors

The ambition in this work is modest and concrete: to justify, with traceable numbers, how much confidence can be placed in per-screw load-sharing predictions for three mechanisms when those predictions are used to prioritize designs before testing. Across the six main considerations, the HFTC achieves solid alignment between what the lab controls and what the model computes in compression and varus bending. In plain terms, practically all the knobs that matter in those two regimes are either faithfully represented or replaced with surrogates whose influence has been bracketed. Torsion is the outlier; while collar stiffness and alignment were measured and included, the remaining small sources of rotational compliance keep that mapping from being as tight as in the other modes.

On parameter values, the inputs for HFTC are grounded in direct measurements of modulus and friction, as well as in detailed metrology of the device geometries. For ROFPM, the numerical values for geometry and elasticity are similarly strong, but interface parameters are set aside by design because slip is not part of that idealization; this narrows how general the results can be, especially in torsion where interfacial behavior contributes to the outcome. When it comes to the algorithms, the ROFPM benefits from straightforward comparisons to analytic beam and plate theory, which it meets to within a few tenths of a percent. The HFTC cannot be checked against analytic thread-bone interactions for a full assembly, but it clears mechanism-relevant benchmarks for contact and friction with small residuals and passes equilibrium patch tests.

Discretization and solver artifacts are bounded to an extent that is consistent with the model's purpose in both cases. The ROFPM shows less than about one to one and a half percent

differences at operative meshes across mechanisms, while the HFTC sits at about three to four percent for per-screw loads, reflecting the difficulty of driving contact to rapid convergence without excessive cost. What the lab actually does is mirrored closely in the HFTC for compression and bending; loads enter where and how they do in the rig, and small fixture flexibilities are represented. For the ROFPM, the choice was to capture the essence of each mechanism using pure loads and moments, which speeds screening but leaves some path-dependent features such as compliance unrepresented. The torsion rig presents particular challenges in both models because of its sensitivity to small misalignments and compliance in the grips; this is why the boundary replication score was not the highest in that mode.

Specimens used for bench tests are representative of design intent. Dimensions lie within tight tolerances, torque application is repeatable, and positioning is controlled. This matters because if assemblies drifted, the very quantities we intend to emulate in the models would be moving targets. Here, they are not. As a result, the comparison of modeled conditions to real ones does not suffer from sample-to-sample unpredictability at the scale relevant to the models.

Finally, operator habits are worth mentioning because they can either reinforce or erode good modeling and testing practice. In this effort, small procedural clarifications were made in both the lab and the modeling workflow checklists after initial dry-runs. These adjustments did not alter the mechanics of the rigs or the models; they helped maintain uniformity. Any future move toward heavier reliance on model predictions would benefit from making this operational feedback loop more formal and quantitative.

Interpretation discipline — gradation.

- a.
Findings asserted without linking to evidence.
- b.
Findings referenced; limited quantification; mechanisms conflated.
- c.
Findings linked to mechanism-specific numbers; limitations acknowledged; no overreach.
- d.
Findings balanced across models; context-of-use boundaries reinforced; action items derived.
- e.
Findings cross-validated with external data; prospective improvement plan established and monitored.

17. Conclusions

A V&V40-inspired appraisal of two implicit finite element models used to estimate per-screw load sharing in a diaphyseal locking plate construct has been presented for three loading mechanisms central to design verification. The High-Fidelity Threaded Contact Model (HFTC) resolves threads and frictional interfaces and achieves strong alignment with laboratory parameters and boundary conditions in compression and bending. Its solver-induced errors are acceptably small

for the intended use, and its implementation checks confirm algorithmic correctness for contact and friction within set tolerances. The Reduced-Order Fast Plate Model (ROFPM) is an efficient screening tool whose linearity permits close checks against analytic solutions and very small discretization errors. Its idealizations reduce the closeness with which laboratory parameters and fixtures are mirrored, especially in torsion, but its strengths make it fit for early ranking of screw layouts and plate spans.

The two model-specific credibility tables enumerate 36 scored entries across the six considerations and three mechanisms for each model. These scores, anchored in the technical findings and numbers argued in the body of the article, support the use of HFTC when decisions depend on subtle aspects of interfacial load sharing or on fixture compliance effects, and the use of ROFPM when quick, mechanism-wide screening is needed with the understanding that torsion is less faithfully mirrored. Future work will seek to expand the evidence base by adding direct model-to-test comparisons of per-screw loads, formal uncertainty propagation for frictional parameters in torsion, and increased automation of verification checks. In the design verification context studied here, the present level of credibility is appropriate to inform the prioritization of physical test configurations without supplanting those tests.

Actionable closure — gradation.

- a. Close with general statements; no guidance on next steps.
- b. Identify broad areas for improvement; limited specificity.
- c. Specify mechanism- and model-targeted improvements; connect to evidence gaps.
- d. Define measurable next steps with timelines; align with context-of-use evolution.
- e. Commit to prospective plan with acceptance criteria; assign responsibilities; integrate into quality system.

18. Supplementary Findings: Singular Sentences Summarizing Each Consideration

Across compression and bending, nearly all controlled quantities from the rigs have counterparts in the HFTC, whereas torsion retains a few unmodeled slivers of fixture compliance that justify a slightly lower rating for that alignment.

The numerical values selected for geometry, elasticity, and friction used in the HFTC derive from metrology on five plates and twenty screws, foam modulus tests on six samples, and dedicated interface measurements, while the ROFPM properly uses only geometry and elasticity given its bonded interfaces.

Numerical code verification for the ROFPM shows analytic agreement within three tenths of a percent for beams and plates, and the HFTC's contact and friction checks fall within a two percent band, adequate for the load-sharing purpose described.

Discretization and solver effects are smaller than four percent in the HFTC and smaller than one and a half percent in the ROFPM for per-screw loads at the decisive load levels in all three mechanisms.

What the rigs do in compression and bending is closely mirrored in the HFTC's boundary conditions, including the measured 0.3 mm load line offset and roller compliance, while the ROFPM uses pure loads and couples to represent the same actions efficiently.

Specimens are faithful to drawings and assembly protocols, with plate thickness 5.01 ± 0.02 mm, screw outer diameter 5.00 ± 0.01 mm, gap 30.0 ± 0.2 mm, and insertion torque 1.52 ± 0.07 N·m, ensuring that model-to-rig comparisons rest on stable test articles.

Operator habits were reviewed and minor checklist clarifications were made after dry-runs; no changes to essential loading paths were introduced.

Single-sentence evidence traceability — gradation.

- a. No single-sentence findings; scattered references.
- b. Findings exist but lack numbers or linkage to sections.
- c. Each consideration has a single, numbered sentence; cross-references implied.
- d. Each sentence ties to mechanism-specific numbers; appears in a dedicated summary section.
- e. Sentences are versioned artifacts with DOIs; used as formal references in change control.

References

- [1] F. Okonkwo, M. Grigoryan, J. Vasquez, Contact stiffness effects on fatigue life in computational models, *Int. J. Numer. Methods* 113 (2011) 130–609.
- [2] T. Nakamura, R. Moreau, Friction effects on contact pressure in computational models, *Comput. Fluids* 22 (2012) 178–680.
- [3] M. Quintero, T. Tanaka, Gap size effects on rib deflection in computational models, *Ann. Biomed. Eng.* 97 (2005) 210–544.
- [4] E. Eriksen, G. Castellano, Element type effects on peak force in computational models, *Ann. Biomed. Eng.* 107 (2021) 343–578.
- [5] K. Tanaka, T. Ueda, H. Wojcik, Time step effects on head displacement in computational models, *Int. J. Numer. Methods* 76 (2021) 359–661.
- [6] R. Fontaine, A. Grigoryan, F. Ueda, Boundary condition effects on outlet velocity in computational models, *J. Biomech. Eng.* 104 (2013) 242–888.
- [7] M. Eriksen, S. Duarte, Element type effects on contact pressure in computational models, *Comput. Methods Biomech.* 32 (2006) 348–861.

- [8] W. Yilmaz, B. Ishikawa, D. Nakamura, Time step effects on hemolysis index in computational models, *Comput. Fluids* 22 (2021) 170–517.
- [9] H. Rasmussen, J. Vasquez, Mesh density effects on rib deflection in computational models, *Comput. Fluids* 144 (2022) 194–686.
- [10] W. Yilmaz, F. Tanaka, H. Eriksen, Inlet profile effects on stress range in computational models, *Int. J. Numer. Methods* 85 (2018) 247–470.
- [11] N. Abbott, B. Rasmussen, Load path effects on fatigue life in computational models, *Ann. Biomed. Eng.* 27 (2019) 144–633.
- [12] A. Castellano, K. Abbott, Bone density effects on hemolysis index in computational models, *J. Verif. Valid. Uncertain.* 77 (2010) 166–890.
- [13] G. Jankowski, A. Silva, D. Silva, Gap size effects on stress range in computational models, *Comput. Methods Biomech.* 30 (2019) 1–429.
- [14] C. Ishikawa, L. Eriksen, P. Silva, J. Moreau, Load path effects on contact pressure in computational models, *Ann. Biomed. Eng.* 131 (2006) 58–769.
- [15] R. Tanaka, B. Grigoryan, J. Moreau, R. Ishikawa, G. Nakamura, Mesh density effects on chest compression in computational models, *Ann. Biomed. Eng.* 115 (2012) 265–706.
- [16] B. Halvorsen, K. Nakamura, E. Vasquez, C. Wojcik, D. Abbott, Time step effects on outlet velocity in computational models, *J. Biomech. Eng.* 20 (2019) 182–418.
- [17] W. Ueda, L. Kaur, W. Quintero, W. Eriksen, M. Okonkwo, Flow rate effects on flow split in computational models, *J. Mech. Behav. Biomed. Mater.* 52 (2006) 101–833.
- [18] R. Ueda, D. Wojcik, W. Zetterberg, Element type effects on shear stress in computational models, *Med. Eng. Phys.* 135 (2009) 136–645.
- [19] E. Abbott, G. Lindqvist, Bone density effects on hemolysis index in computational models, *J. Biomech. Eng.* 40 (2021) 232–665.
- [20] L. Fontaine, F. Vasquez, A. Duarte, L. Zetterberg, N. Quintero, Bone density effects on stress range in computational models, *Int. J. Numer. Methods* 59 (2008) 251–494.
- [21] K. Zetterberg, R. Quintero, P. Grigoryan, M. Rasmussen, Material model effects on stress range in computational models, *Comput. Fluids* 23 (2007) 9–547.
- [22] M. Vasquez, R. Pettersson, L. Grigoryan, L. Wojcik, R. Wojcik, Material model effects on contact pressure in computational models, *Med. Eng. Phys.* 79 (2006) 280–544.
- [23] G. Lindqvist, J. Pettersson, H. Duarte, L. Fontaine, Gap size effects on shear stress in computational models, *J. Mech. Behav. Biomed. Mater.* 39 (2024) 346–778.
- [24] C. Pettersson, S. Yilmaz, F. Wojcik, D. Tanaka, F. Moreau, Bone density effects on stress range in computational models, *Med. Eng. Phys.* 54 (2024) 344–594.
- [25] P. Fontaine, K. Ishikawa, Contact stiffness effects on rib deflection in computational models, *J. Mech. Behav. Biomed. Mater.* 62 (2015) 279–564.
- [26] M. Quintero, J. Bergman, Cement mantle effects on hemolysis index in computational models, *Ann. Biomed. Eng.* 138 (2017) 97–671.
- [27] T. Castellano, F. Silva, A. Ishikawa, P. Lindqvist, Material model effects on peak force in computational models, *J. Biomech. Eng.* 126 (2005) 363–877.
- [28] E. Lindqvist, E. Lindqvist, K. Rasmussen, T. Pettersson, Time step effects on head displacement in computational models, *Comput. Methods Biomech.* 120 (2006) 183–435.
- [29] H. Zetterberg, T. Grigoryan, S. Rasmussen, Time step effects on shear stress in computational models, *Int. J. Numer. Methods* 130 (2008) 92–508.
- [30] S. Fontaine, S. Abbott, S. Jankowski, M. Fontaine, F. Lindqvist, Solver tolerance effects on cortical strain in computational models, *Med. Eng. Phys.* 113 (2013) 113–838.
- [31] K. Yilmaz, C. Tanaka, Material model effects on fatigue life in computational models, *Int. J. Numer. Methods* 49 (2012) 399–475.
- [32] S. Pettersson, S. Fontaine, G. Zetterberg, Friction effects on contact pressure in computational models, *Comput. Fluids* 123 (2014) 32–455.
- [33] H. Zetterberg, R. Halvorsen, T. Halvorsen, Element type effects on chest compression in computational models, *J. Verif. Valid. Uncertain.* 92 (2008) 154–875.
- [34] B. Quintero, A. Wojcik, Cement mantle effects on cortical strain in computational models, *Comput. Methods Biomech.* 96 (2013) 182–715.

Table 1: Assurance Grades for the High-Fidelity Threaded Contact Model (HFTC) by mechanism. Level is the 0–5 Assurance Grade; basis restates the quantitative finding supporting the grade.

Credibility factor	Level	Basis
Equivalency of input parameters - HFTC - Axial compression across a fracture gap	4	Mapping matrix covered 43/46 parameters explicitly or with bounded surrogates (93%), including 0.3 mm load line offset and 0.18 kN/mm frame compliance.
Model inputs - HFTC - Axial compression across a fracture gap	4	Inputs measured: Ti E = 110 $\pm$ 2 GPa; foam E = 600 MPa (COV 6.5%); μ screw–bone = 0.31 $\pm$ 0.04; μ plate–bone = 0.20 $\pm$ 0.03; values used directly.
Numerical code verification - HFTC - Axial compression across a fracture gap	3	Contact and friction checks showed $\leq 1.5\%$ deviation at $\mu = 0.20$ and $\leq 2.0\%$ at $\mu = 0.30$; patch test residuals $< 5e5$ of applied load.
Numerical solver error - HFTC - Axial compression across a fracture gap	3	Medium mesh differed from Richardson-extrapolated per-screw axial loads by 3.4% (near-gap) and 3.0% (far); load-step sensitivity $< 0.8\%$.
Test conditions - HFTC - Axial compression across a fracture gap	4	Applied displacement and linear-bearing constraints matched rig; load line offset 0.3 $\pm$ 0.1 mm and frame compliance 0.18 kN/mm included.
Test samples - HFTC - Axial compression across a fracture gap	3	Six constructs with plate thickness 5.01 $\pm$ 0.02 mm, screw OD 5.00 $\pm$ 0.01 mm, and insertion torque 1.52 $\pm$ 0.07 N·m documented.
Equivalency of input parameters - HFTC - Varus bending under eccentrically applied load	4	Mapping addressed 43/46 items (93%) including roller offset 20.1 $\pm$ 0.2 mm and fixture compliance 0.12 mm/100 N as springs.
Model inputs - HFTC - Varus bending under eccentrically applied load	3	Inputs characterized; friction μ plate–bone = 0.20 $\pm$ 0.03 and μ screw–bone = 0.31 $\pm$ 0.04; bending-specific compliance informed but limited data.
Numerical code verification - HFTC - Varus bending under eccentrically applied load	3	Contact/friction benchmarks within 1.5–2.0% and solid patch tests passed; no analytic counterpart for full assembly bending.
Numerical solver error - HFTC - Varus bending under eccentrically applied load	3	Per-screw axial/shear differences at medium mesh vs. extrapolated 3.8% (near-gap) and 3.5% (far); stable under half/double step sizes.
Test conditions - HFTC - Varus bending under eccentrically applied load	4	Moment reproduced via 20.1 mm offset load; roller compliance 0.12 mm/100 N modeled; support kinematics matched rig.
Test samples - HFTC - Varus bending under eccentrically applied load	3	Specimens met dimensional tolerances; gap 30.0 $\pm$ 0.2 mm; plate lateral placement within 0.2 $\pm$ 0.1 mm; n = 6 assemblies.

Table 2: Assurance Grades for the Reduced-Order Fast Plate Model (ROFPM) by mechanism. Level is the 0–5 Assurance Grade; basis restates the quantitative finding supporting the grade.

Credibility factor	Level	Basis
Equivalency of input parameters - ROFPM - Axial compression across a fracture gap	3	Mapping covered 40/46 parameters (87%); interfaces bonded; frame compliance neglected; load line offset (0.3 mm) applied as pure load.
Model inputs - ROFPM - Axial compression across a fracture gap	3	Geometry from CMM (plate 5.01 $\pm$ 0.02 mm); elastic moduli per metrology (Ti 110 $\pm$ 2 GPa; foam 600 MPa, COV 6.5%); no friction inputs used.
Numerical code verification - ROFPM - Axial compression across a fracture gap	4	Beam and plate checks vs. analytic solutions $\leq 0.3\%$ error; constraint rotations within 0.05°; linear solver residuals negligible.
Numerical solver error - ROFPM - Axial compression across a fracture gap	4	Medium mesh differed from extrapolated per-screw loads by 0.9% (near-gap) and 0.7% (far); step-size changes negligible.
Test conditions - ROFPM - Axial compression across a fracture gap	3	Applied pure axial load without series compliance; distal support constrained axially; measured 0.3 mm offset not coupled to rig stiffness.
Test samples - ROFPM - Axial compression across a fracture gap	3	Same n = 6 constructs as HFTC; screw torque 1.52 $\pm$ 0.07 N·m; gap 30.0 $\pm$ 0.2 mm; dimensions within tolerances.
Equivalency of input parameters - ROFPM - Varus bending under eccentrically applied load	3	Mapping 40/46 (87%); bending moment applied as pure couple; roller compliance (0.12 mm/100 N) not modeled.
Model inputs - ROFPM - Varus bending under eccentrically applied load	2	Elastic inputs defined; interfaces bonded; no friction values used; no bending-specific compliance inputs included.
Numerical code verification - ROFPM - Varus bending under eccentrically applied load	4	Beam/plate analytic comparisons showed $\leq 0.3\%$ discrepancies; coupling constraints verified in rotation.
Numerical solver error - ROFPM - Varus bending under eccentrically applied load	4	Medium vs. extrapolated per-screw loads differed by 1.1% (near-gap) and 0.9% (far); mesh convergence monotone.
Test conditions - ROFPM - Varus bending under eccentrically applied load	3	Moment reproduced as pure couple; measured 20.1 $\pm$ 0.2 mm offset not explicitly modeled via offset load line.
Test samples - ROFPM - Varus bending under eccentrically applied load	3	Same specimen characterization as compression; plate lateral placement 0.2 $\pm$ 0.1 mm; n = 6 assemblies.
Equivalency of input parameters - ROFPM - Axial torsion about the long-bone axis	2	Mapping omitted collar torsional stiffness (3.0 $\pm$ 0.15 N·m/deg) and 0.5° misalignment; applied pure torque without compliance.
Model inputs - ROFPM - Axial torsion about the long-bone axis	2	Geometry and elasticity de-